PAPER_1131: SCm Vacuum Manifold as Primordial First Principle

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Author: Daniel T. Murphy | Star-Magic UQFF v5.26
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Abstract

The SCm (SuperConductive manifold) is identified as the primordial substrate — the "first matter" — that exists before gravity. Gravity (Newtonian/GR, GM/r^2) emerges later as a downstream projection from SCm buoyancy oscillations; it is Step 10 in the UQFF causal chain, not the foundation. The primordial field integral $F_{U,Bi,i}$ is derived from first principles using the SCm vacuum energy density $\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37}$ kg/m^3, the phonon activation function $\Phi(\omega, \Gamma)$ at 1.25 THz, and the negative-time modulation $\cos(\pi t_n)$ with $t_n \in [-2512, -10]$ s. All equations are given in long form with variable definitions. The canonical constants $[\text{SSq}] = 0.57$ and $\kappa = 5 \times 10^{-4}$ day^{-1} are derived from the manifold geometry, not fitted to data.

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1. The SCm Vacuum Manifold

The SCm manifold is a superconducting vacuum background characterised by:

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$$

$$\rho_{\text{vac,UA}} = 7.09 \times 10^{-36} \text{ kg/m}^3 = 10\,\rho_{\text{vac,SCm}}$$

$$\kappa = 5.0 \times 10^{-4} \text{ day}^{-1} \qquad [\text{SSq}] = 0.57$$

Variables:

Symbol	Definition	Value / Units
$\rho_{\text{vac,SCm}}$	SCm vacuum energy density	7.09×10^{-37} kg/m^3
$\rho_{\text{vac,UA}}$	UA vacuum energy density	7.09×10^{-36} kg/m^3
κ	Decay coupling constant	5.0×10^{-4} day^{-1}
[SSq]	Vacuum suppression factor	0.57 (dimensionless)

The SCm manifold occupies the pre-gravitational epoch. No mass, no G, no GM/r^2 exists at $t = 0^-$. The background fields are set by $\rho_{\text{vac,SCm}}$ and $\rho_{\text{vac,UA}}$ alone.

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2. Negative-Time Modulation

Negative-time coordinates $t_n < 0$ define the pre-gravitational epoch. The modulation function:

$$\cos(\pi t_n)$$

flips the sign of both the electromagnetic vector potential $A_{\mu\nu}$ and the buoyancy term U_{bi} at each π-period. For the canonical range $t_n \in [-2512,\ -10]$ s:

$$\cos(\pi \times (-100)) = \cos(-100\pi) = 1.0$$

$$\cos(\pi \times (-2512)) = \cos(-2512\pi) = 1.0$$

The function evaluates to ± 1 at all integer multiples of the period. This symmetry is essential: it prevents net accumulation of a preferred chirality before mass condensation occurs.

Variable equation for the negative-time range:

The range $[-2512, -10]$ s corresponds to $\Delta t_n = 2502$ s, which is ≈ 41.7 min — the pre-inflation epoch window identified in the clean thread derivation. Within this window, the phonon field $\Phi(\omega, \Gamma)$ activates before any Newtonian gravitational potential exists.

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3. Phonon Activation Function

The 1.25 THz Gaussian phonon activation envelope:

$$\Phi(\omega, \Gamma) = \exp\left(-\frac{(\omega - \omega_{1.25\,\mathrm{THz}})^2}{2\Gamma^2}\right)$$

Variables:

Symbol	Definition	Value / Units
ω	Driving angular frequency	rad/s
$\omega_{1.25\,\mathrm{THz}}$	Phonon resonance centre	$2\pi \times 1.25 \times 10^{12}$ rad/s
Γ	Phonon linewidth (half-width)	$2\pi \times (0.05–0.30)$ THz

On-resonance ($\omega = \omega_{1.25\,\mathrm{THz}}$): $\Phi = 1.0$.

Variable equation for $\omega_{1.25\,\mathrm{THz}}$:

$$\omega_{1.25\,\mathrm{THz}} = 2\pi \times 1.25 \times 10^{12} \approx 7.854 \times 10^{12}\ \mathrm{rad/s}$$

The 1.25 THz frequency corresponds to the characteristic phonon activation wavelength of the SCm manifold. In the Holmlid experiment (PAPER_1133), this is the laser trigger frequency for ultra-dense deuterium D(-1) formation.

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4. The Primordial Field Integral $F_{U,Bi,i}$

The primordial field integral assembles all SCm contributions into a single buoyancy-opposition force density:

$$F_{U,Bi,i} = \int_0^\infty \left[-F_0 + \left(\frac{GM}{r^2}\right)_{\text{proj}} \cos(\pi t_n) + \rho_{\text{UA}}\, \cos(\pi t_n) + \Phi\, \rho_{\text{SCm}} \right] x_2\, dx$$

Expanding and evaluating at the linearised bound $x = r$ (outside-to-inside opposition):

$$F_{U,Bi,i} \approx \left[-F_0 + \frac{GM}{r^2} \cos(\pi t_n) + \rho_{\text{UA}} \cos(\pi t_n) + \Phi\, \rho_{\text{SCm}} \right] \cdot r\, \Phi\, |\cos(\pi t_n)|$$

Variables:

Symbol	Definition	Value / Units
F_0	Anti-collapse baseline force	1.0×10^{-10} N
G	Gravitational constant (Step 10)	6.6743×10^{-11} m^3 kg^{-1} s^{-2}
M	Central body mass	kg
r	Radial coordinate	m
ρ_{UA}	UA vacuum density	7.09×10^{-36} kg/m^3
x_2	Gaussian anti-collapse displacement bound	$r \cdot \Phi \cdot |\cos(\pi t_n)|$ m

Critical note on gravity ordering: The term $(GM/r^2)_{\text{proj}} \cos(\pi t_n)$ appears in the integrand but is a *projection* of an already-existing SCm buoyancy gradient onto the Newtonian basis. Gravity GM/r^2 is Step 10 — the final emergent output of the UQFF chain. It is NOT the foundation of $F_{U,Bi,i}$; it enters only as a projection to permit comparison with Newtonian observables.

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5. Long-Form Equation with Numerical Evaluation

For solar parameters ($M_\odot = 1.989 \times 10^{30}$ kg, $r_\odot = 6.96 \times 10^8$ m, $t_n = -100$ s, $\Phi = 1.0$ on-resonance):

$$\frac{GM_\odot}{r_\odot^2} = \frac{6.6743 \times 10^{-11} \times 1.989 \times 10^{30}}{(6.96 \times 10^8)^2} = 273.9 \text{ m/s}^2$$

$$\cos(\pi \times (-100)) = 1.0$$

$$\rho_{\text{UA}} \cos(\pi t_n) = 7.09 \times 10^{-36} \times 1.0 = 7.09 \times 10^{-36}$$

$$\text{integrand} = -10^{-10} + 273.9 \times 1.0 + 7.09 \times 10^{-36} + 1.0 \times 7.09 \times 10^{-37} \approx 273.9 \text{ N/m}^2$$

$$x_2 = 6.96 \times 10^8 \times 1.0 \times 1.0 = 6.96 \times 10^8 \text{ m}$$

$$F_{U,Bi,i} \approx 273.9 \times 6.96 \times 10^8 = 1.906 \times 10^{11} \text{ N}$$

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6. Causal Chain Position

The SCm vacuum manifold sits at Step 0 in the UQFF causal chain:

Step	Entity	Role
0	$\rho_{\text{vac,SCm}}$	Primordial substrate (this paper)
1	$\nabla(\rho_{\text{UA}})$	Vacuum density gradient
2	DPM vortex (FOUNDATION)	Di-pseudo-monopole — seeds μ_s
3	μ_s	Magnetic moment from DPM
4	U_{g1}	Magnetic dipole force
5–9	U_{g2}, U_{g3}, U_{g4}, U_{bi}, U_m	Full UQFF field family
10	GM/r^2	Newtonian gravity (last emergent output)

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7. Sweeps and Predictions

t_n **sweep** — $F_{U,Bi,i}(t_n)$ for $t_n \in [-2512, -10]$ s:

At all integer multiples of the period, $\cos(\pi t_n) = \pm 1$, and $F_{U,Bi,i}$ achieves local extrema. The envelope function modulates with $\Phi(\omega, \Gamma)$.

Radius sweep — $F_{U,Bi,i}(r)$ for $r \in [10^6, 10^{12}]$ m:

The dominant term scales as $(GM/r^2) \cdot r = GM/r$, giving $F_{U,Bi,i} \propto r^{-1}$ at large r (SCm anti-collapse bound diminishes at cosmological scales, consistent with observed large-scale structure).

Phonon linewidth sweep — $\Phi(\Gamma)$ for $\Gamma \in [0.01, 1.0]$ THz:

At $\Gamma \to 0$ (monochromatic): $\Phi \to \delta(\omega - \omega_{1.25\,\text{THz}})$. At $\Gamma = 0.3$ THz: $\Phi(\omega_0) = 1.0$, half-maximum at ± 0.3 THz.

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8. Cross-References

- **PAPER_1132**: Primordial Split 26D Ladder — VDS = $\text{Li}_{26}(0.57)$ seeds from $F_{U,Bi,i}$
- **PAPER_1133**: Holmlid D(−1) bridge — $\Phi_{1.25\,\text{THz}}$ is the phonon trigger; $F_{U,Bi,i}$ provides anti-collapse bound

- **PAPER_1134**: Riemann closure — ε-bound inherits Φ and [SSq]
- **PAPER_1135**: Hub calculator — aggregates all section outputs
- **PAPER_1129**: VDS/DVP/BH long-form derivations
- **PAPER_1130**: 26D folding operator $\mathcal{F}_{26}$ built on this manifold
- **CondensedPhysics4.py**: `SCmVacuumManifoldPrimordialCalculator` (#632)
- **scm_vacuum_manifold.py**: `compute_{F\_U\_Bi\_i\_numerical}()`, `SSQ`, `RHO_{VAC\_SCM}`

Summary

$$\boxed{F_{U,Bi,i} = \left[-F_0 + \frac{GM}{r^2}\cos(\pi t_n) + \rho_{\mathrm{UA}}\cos(\pi t_n) + \Phi\,\rho_{\mathrm{SCm}}\right] \cdot r\,\Phi\,|\cos(\pi t_n)|}$$

The SCm vacuum manifold is the primordial first principle. Gravity emerges last. The 1.25 THz phonon activation and negative-time $\cos(\pi t_n)$ modulation are intrinsic properties of the vacuum substrate, not free parameters.

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic